

Stereoselective Synthesis of β -Branched Baylis–Hillman Adducts via Organozinc Species

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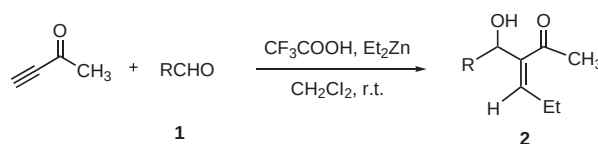
Abstract: An efficient one-pot, three-component coupling reaction for the synthesis of unusual β -branched Baylis–Hillman adducts has been developed. Organozinc species $\text{CF}_3\text{CO}_2\text{ZnEt}$ added to α,β -acetylenic ketones in a 1,4-fashion to yield allenolates, which reacted with aldehydes providing highly functionalized trisubstituted olefins in good to excellent yields with stereoselectivity.

Key words: Baylis–Hillman adducts, organozinc species, acetylenic ketones, aldehydes

The Baylis–Hillman reaction is an efficient carbon–carbon bond-forming reaction in organic synthesis.^{1,2} β -Substituted Baylis–Hillman adducts as highly functionalized trisubstituted olefins can be subjected to subsequent transformations for the synthesis of many natural products and synthetic derivatives.³ However, the β -substituted olefinic substrates do not normally undergo the Baylis–Hillman reaction,⁴ so alternative methods, such as the reaction of copper,⁵ aluminum,^{6,7} and boron reagents⁸ with α,β -acetylenic carbonyl compounds, have been developed to overcome the limitation of the original Baylis–Hillman reaction.

In 1983, Marino and coworkers reported the stereospecific coupling reaction of (α -carbethoxyvinyl) cuprate reagents with ketones, but mixtures of isomers were obtained when aldehydes were used as the electrophilic acceptors.^{5c} Recently, vinyl aluminum species have been reported to afford a general synthesis of β -substituted Baylis–Hillman adducts by coupling with various electrophiles such as aldehydes, fluoro ketones, α -keto esters, α -acyl cyanides, and α,β -acetylenic ketones.⁷ The vinyl aluminum species were prepared from reductive addition of diisobutylaluminum hydride (DIBALH) to α,β -acetylenic carbonyl compounds in the presence of hexamethylphosphoric amide or 4-methylmorpholine. In 1999, Kabalka and coworkers successfully synthesized stereodefined functionalized trisubstituted olefins via the self-condensation reaction of α,β -acetylenic ketones in the presence of dicyclohexylborane. This reaction is believed to proceed through allenoxyborinate formed by enolization of vinylborane intermediate. The specially hindered allenoxyborinate can also be trapped by aldehydes and ketones to produce *Z*-trisubstituted olefins.^{8b}

Although organozinc reagents have been widely used in the conjugate addition to acetylenic esters,^{9,10} little attention has been given to the reactivity of organozinc reagents with α,β -acetylenic ketones.¹¹ Our continuing interest in exploring the reactivity and behavior of the organozinc species $\text{CF}_3\text{CO}_2\text{ZnR}$ with electrophiles¹² has led us to find that $\text{CF}_3\text{CO}_2\text{ZnEt}$ can react with α,β -acetylenic ketones as Michael-type addition in the absence of a copper catalyst. Here, we wish to report this new procedure for stereoselective synthesis of β -branched Baylis–Hillman adducts by a one-pot, three-component coupling reaction of α,β -acetylenic ketones with organozinc species and aldehydes (Scheme 1).



Scheme 1

Organozinc species $\text{CF}_3\text{CO}_2\text{ZnEt}$ can be readily prepared by stirring ZnEt_2 with CF_3COOH in CH_2Cl_2 at 0 °C for 30 minutes. We treated the in situ generated $\text{CF}_3\text{CO}_2\text{ZnEt}$ with 3-butyne-2-one and 4-methoxybenzaldehyde at room temperature for three hours and found that β -substituted Baylis–Hillman adduct **2a** was formed in 74% isolated yield. In an effort to optimize the reaction conditions, the effects of reaction temperature, solvent, and organozinc species were studied. The reaction produced **2a** in 86% yield after five hours at room temperature but required ten hours for obtaining 78% at 0 °C. Thus, room temperature was chosen for further studies. The studies revealed that CH_2Cl_2 was suitable solvent. Toluene, diethyl ether, and THF gave a low yield of 36%, 7% and 11%, respectively, within a five-hour reaction period. CF_3COOH was also crucial for this reaction. The organozinc species $\text{Cl}_3\text{CO}_2\text{ZnEt}$ and ZnEt_2 , in place of $\text{CF}_3\text{CO}_2\text{ZnEt}$, provided **2a** with low yields of 75% and 17%, respectively.

With the optimal reaction conditions at hand, a variety of aldehydes were then allowed to react with $\text{CF}_3\text{CO}_2\text{ZnEt}$ and 3-butyne-2-one. These results were presented in Table 1.¹³ Both aromatic and aliphatic aldehydes can be employed as suitable electrophilic acceptors. For aromatic aldehydes, substitution of an electron-withdrawing group or an electron-donating group on the aromatic ring resulted in no obvious effect on the reaction yields. However,

Table 1 Reaction of 3-Butyn-2-one with $\text{CF}_3\text{COOZnEt}$ and Aldehydes^a

Entry	R	Time (h)	Product	Yield (%) ^b
1	4-MeOC ₆ H ₄	3	2a	74
2	4-MeOC ₆ H ₄	5	2a	86
3 ^c	4-MeOC ₆ H ₄	5	2a	35
4 ^c	4-MeOC ₆ H ₄	10	2a	78
5 ^d	4-MeOC ₆ H ₄	6	2a	75
6 ^e	4-MeOC ₆ H ₄	5	2a	17
7	3-MeOC ₆ H ₄	6	2b	95
8	C ₆ H ₅	5	2c	88
9	4-ClC ₆ H ₄	5	2d	89
10	2-ClC ₆ H ₄	5	2e	83
11	4-NO ₂ C ₆ H ₄	5	2f	31
12 ^f	4-NO ₂ C ₆ H ₄	7	2f	79
13	4-MeC ₆ H ₄	6	2g	84
14	1-Naphthyl	5	2h	81
15	C ₆ H ₅ CH ₂ CH ₂ CH ₂	5	2i	78
16	<i>trans</i> -C ₆ H ₅ CH=CH	6	2j	82
17	CH ₃ CH ₂ CH ₂ CH ₂	5	2k	81

^a In a typical reaction, aldehyde (0.5 mmol) was added to a solution of $\text{CF}_3\text{CO}_2\text{ZnEt}$, from ZnEt_2 (0.75 mmol) and CF_3COOH (0.75 mmol), then followed by 3-butyne-2-one (0.75 mmol).

^b Yields after purification by column chromatography.

^c The reaction was carried out at 0 °C.

^d CCl_3COOH was used, in place of CF_3COOH .

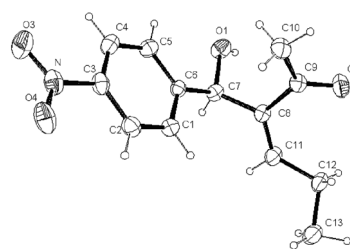
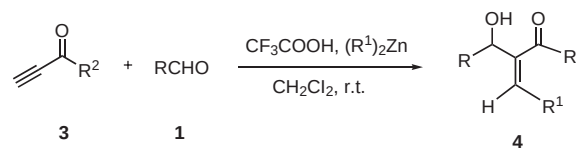
^e No CF_3COOH was added.

^f Performed at -78 °C for 1 h, then at 0 °C for 6 h.

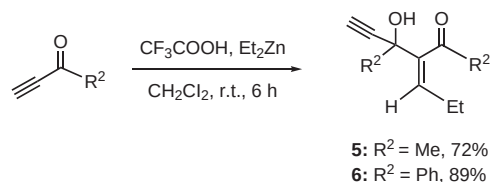
4-nitrobenzaldehyde provided the desired product **2f** in a very poor yield of 31% at room temperature. Through analysis of the reaction products, we found that the high reactivity of the electrophilic acceptor led to direct alkylation of $\text{CF}_3\text{CO}_2\text{ZnEt}$ with 4-nitrobenzaldehyde to produce 1-(4-nitrophenyl)-1-propanol. Fortunately, when the reaction was performed at -78 °C for one hour, then at 0 °C for another six hours before it was quenched, we got product **2f** in a good yield of 79%.

Interestingly, the stereochemistry of the β -branched Baylis–Hillman adducts was determined to be the *Z*-configuration for all cases in our new reaction system as revealed by ^1H NMR analyses of the crude products. The *E*-isomer of the product was not observed from the ^1H NMR and ^{13}C NMR spectra. Most of the products in Table 1 are oils except product **2f**, which was obtained as a solid product, so we were able to obtain the X-ray crystal structure of **2f**. As shown in Figure 1, its relative configuration was further confirmed as the *Z*-configuration.¹⁴

This mild methodology can also be extended to aromatic α,β -acetylenic ketones as the conjugated addition acceptors (Scheme 2, Table 2). For example, the reaction of 1-phenylprop-2-yn-1-one with $\text{CF}_3\text{CO}_2\text{ZnEt}$ and 4-methoxybenzaldehyde provided the corresponding product in high yield (Table 2, entry 2). Substitution of an electron-donating group on the aromatic ring provided a little lower yield than that of an electron-withdrawing group on the aromatic ring (Table 2, entries 6, 7). Organozinc species formed from diphenylzinc, in place of diethylzinc, was found to provide the corresponding adducts with low yields. Furthermore, reactions with 3-butyne-2-one gave lower yield than that of 1-phenylprop-2-yn-1-one (Table 2, entries 13, 14), which might be ascribed to the lower reactivity of alkylic α,β -acetylenic ketones than that of aromatic α,β -acetylenic ketones. ^1H NMR analyses of the crude products indicated that no formation of the corresponding *E*-isomers occurred in these reactions.

**Figure 1** X-ray structure of the product **2f****Scheme 2**

In addition to the expected products, a small amount (<5%) of self-condensation products was also obtained as by-product via the reaction of allenolate zinc intermediate with the starting α,β -acetylenic ketones. In the absence of aldehyde, the reaction of $\text{CF}_3\text{CO}_2\text{ZnEt}$ (0.6 mmol) with 3-butyne-2-one (1.0 mmol) and 1-phenylprop-2-yn-1-one (1.0 mmol) gave the desired self-condensation products with 72% and 89% yields, respectively (Scheme 3).

**Scheme 3**

Considering a cyclic transition state model proposed by Taniguchi and Kishi,^{15,16} we propose a mechanism depicted in Scheme 4 for the reaction. The initial reaction step involves organozinc species XZnR^1 ($\text{X} = \text{Et}, \text{CCl}_3\text{CO}_2$,

Table 2 Reaction of α,β -Acetylenic Ketones with $\text{CF}_3\text{CO}_2\text{ZnR}^1$ and Aldehydes^a

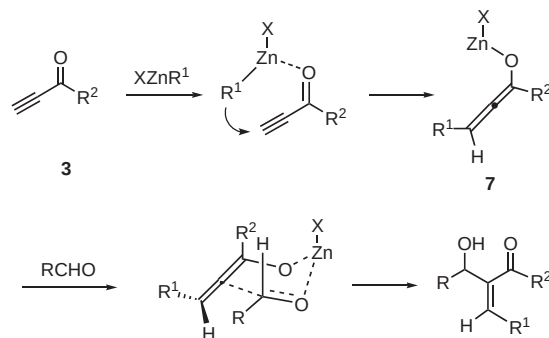
Entry	R ¹	R ²	R	Product	Yield (%) ^b
1	Et	C ₆ H ₅	C ₆ H ₅	4a	91
2	Et	C ₆ H ₅	4-MeOC ₆ H ₄	4b	89
3	Et	C ₆ H ₅	4-ClC ₆ H ₄	4c	91
4	Et	C ₆ H ₅	4-MeC ₆ H ₄	4d	90
5	Et	4-ClC ₆ H ₄	4-MeOC ₆ H ₄	4e	93
6	Et	4-ClC ₆ H ₄	C ₆ H ₅	4f	85
7	Et	4-MeOC ₆ H ₄	C ₆ H ₅	4g	77
8	Et	4-MeOC ₆ H ₄	4-ClC ₆ H ₄	4h	74
9	Et	C ₆ H ₅ CH ₂ CH ₂	4-MeOC ₆ H ₄	4i	84
10	Et	C ₆ H ₅ CH ₂ CH ₂	C ₆ H ₅	4j	84
11	Et	C ₆ H ₅ CH ₂ CH ₂	4-ClC ₆ H ₄	4k	86
12	C ₆ H ₅	C ₆ H ₅	C ₆ H ₅	4l	50
13	C ₆ H ₅	C ₆ H ₅	4-ClC ₆ H ₄	4m	54
14	C ₆ H ₅	Me	4-ClC ₆ H ₄	4n	29

^a In a typical reaction, (R¹)₂Zn (0.75 mmol), CF₃COOH (0.75 mmol), acetylenic ketones (0.75 mmol), and aldehydes (0.5 mmol).

^b Yields after purification by column chromatography.

CF₃CO₂) reacting with α,β -acetylenic ketone through a Michael-type addition to form an active β -substituted allenolate zinc intermediate **7**. The formation of this intermediate is accelerated by the coordination of carbonyl oxygen of acetylenic ketone to the Lewis acid zinc atom. So organozinc species XZnR^1 with strong Lewis acid character will give allenolate **7** smoothly, and providing high product yield. Actually, the reaction is drastically affected by the X group of XZnR^1 . On the other hand, R² group of acetylenic ketone also has an influence on the formation of allenolate intermediate **7**. As shown in Table 2, electron-withdrawing R² group will accelerate its formation and give high yield. The second reaction step proceeds through aldehyde carbonyl oxygen complexation to zinc, which serves to orient the carbonyl group of aldehyde for attack on the allenolate. At this point, the stereochemistry of the reaction is governed by steric interaction between aldehyde R group and allenolate. The reaction occurred exclusively from the less hindered side of the allenolate **7**, producing the sole Z-olefin.

In summary, a mild synthetic method for Z- β -substituted Baylis–Hillman-type adducts has been developed. This new reaction system provides functionalized trisubstituted olefins with good chemical yields and stereoselectivity.

**Scheme 4**

Acknowledgment

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- (13) **Typical Procedure for Reaction of α,β -Acetylenic Ketones with $\text{CF}_3\text{CO}_2\text{ZnEt}$ and Aldehydes.**
To a solution of Et_2Zn (78.5 μL , 0.75 mmol) in 2 mL CH_2Cl_2 at 0 °C was added dropwise CF_3COOH (58 μL , 0.75 mmol) slowly via syringe under N_2 . After stirring for 30 min at 0 °C, aldehyde (0.5 mmol) was added, and then acetylenic ketone (0.75 mmol) was added. The mixture was stirred for 5 h at r.t. until TLC indicated complete consumption of the starting aldehyde. The reaction was quenched by sat. aq NH_4Cl and extracted with Et_2O (3×10 mL). The combined organic extracts was washed with brine, dried over MgSO_4 and concentrated under reduced pressure to an oil residue. The desired product was isolated by silica gel chromatography with petroleum ether– EtOAc (10:1 to 5:1). Data for compound **2a**: ^1H NMR (300 MHz, CDCl_3): δ = 7.18 (d, J = 8.5 Hz, 2 H), 6.80 (d, J = 8.5 Hz, 2 H), 5.80 (t, J = 7.5 Hz, 1 H), 5.32 (d, J = 4.9 Hz, 1 H), 3.72 (s, 3 H), 2.75 (d, J = 4.9 Hz, 1 H), 2.22 (dq J = 7.5, 7.5 Hz, 2 H), 2.10 (s, 3 H), 0.99 (t, J = 7.5 Hz, 3 H) ppm. ^{13}C NMR (75 MHz, CDCl_3): δ = 204.63, 159.18, 142.98, 138.67, 133.96, 127.81, 113.92, 75.07, 55.31, 31.67, 22.87, 14.01 ppm. IR (neat): ν = 3435, 2966, 2838, 1683, 1611, 1585, 1512, 1249, 1174, 1033, 837 cm^{-1} . HRMS (CI): m/z calcd for $\text{C}_{14}\text{H}_{19}\text{O}_3$ [MH^+]: 235.1334; found: 235.1336.
- (14) Crystal data of **2f**: $\text{C}_{13}\text{H}_{15}\text{NO}_4$, crystal system, orthorhombic; space group, Pbca ; unit cell dimensions, a = 11.403 (2) Å, α = 90°; b = 8.844 (1) Å, β = 90°; c = 25.584 (4) Å, γ = 90°; volume, Z = 2580.14 (52) Å³; D_{cald} = 1.283 g/cm³; Z = 8; μ = 0.096 mm⁻¹; F_{000} = 1056; full-matrix least-squares refinement on F^2 ; final R indices [$I > 2\sigma(I)$], $R1$ = 0.0385, $wR2$ = 0.0831; R indices (all data), $R1$ = 0.0922, $wR2$ = 0.0927.
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